

CSE 419: ARTIFICIAL INTELLIGENCE

TERM PROJECT #2 REPORT

Multi-Agent Coordination System via First-Order Logic (FOL) for Disaster Rescue Operations

1. INTRODUCTION & PROBLEM FORMULATION

In disaster relief operations, efficient resource allocation and task scheduling are critical to minimizing human casualties and infrastructural damage. This project implements a Knowledge-Based Multi-Agent System (MAS) that handles spatial coordination and assignment of diverse rescue robots to critical tasks.

The system operates over a structured environment model defined by tasks, robots, and spatial locations. The objective is to find a rational task assignment plan that satisfies all strict constraints while minimizing the global operational cost.

2. KNOWLEDGE REPRESENTATION VIA FIRST-ORDER LOGIC (FOL)

The environment entities and operational conditions are modeled using First-Order Logic (FOL) predicates and strict axioms.

FOL Predicates:

- Task(t): t is a valid disaster task.
- Robot(r): r is an active rescue robot.
- Type(t, type): Operational type of task t.
- Priority(t, p): Priority level of task t.
- Location(e, l): Spatial coordinate of entity e.
- Capability(r, c): Capability of robot r.
- Assigned(t, r): Task assignment relation.
- Compatible(type, c): Compatibility relation.

Hard Constraints:

1. Every task must be assigned to exactly one robot (Exact Assignment / Uniqueness):

$$\forall t (\text{Task}(t) \rightarrow \exists! r \text{ Assigned}(t,r))$$

2. Robots can only execute compatible tasks (Capability Matching):

$$\forall t,r (\text{Assigned}(t,r) \rightarrow \text{Compatible}(\text{Type}(t), \text{Capability}(r)))$$

3. Higher-priority tasks must be processed before lower-priority tasks (Priority Ordering):

$$\forall t_1, t_2 (\text{Priority}(t_1) > \text{Priority}(t_2) \rightarrow \text{ScheduledBefore}(t_1, t_2))$$

3. SYSTEM ARCHITECTURE & MULTI-AGENT DESIGN

The system consists of three cooperating modules:

1. Environment Model:

Initializes the map, robot profiles, coordinates, and task definitions.

2. Assignment Agent:

Explores valid assignments, computes heuristic costs, and selects the optimal plan.

3. Critic Agent:

Checks logical consistency and validates all FOL rules.

4. HEURISTIC COST FUNCTION & OPTIMIZATION

To achieve computational rationality under strict constraint satisfaction, the system implements a formal heuristic cost model based on spatial travel times.

Efficiency Heuristic Function:

The operational cost for a specific robot (r) to execute a designated task (t) is directly proportional to their spatial distance and inversely proportional to the robot's operational speed:

$$\text{Cost}(r,t) = \text{Distance}(r,t) / \text{Speed}(r)$$

Global Optimization Goal (Constraint Minimization):

The Assignment Agent explores the state-space matrix to determine a valid configuration plan. The ultimate goal of the heuristic search is to minimize the cumulative global cost of all assignments:

$$\text{Minimize: } \sum \text{Cost}(r,t)$$

Where $\text{Distance}(r,t)$ represents the spatial metric between the **current (updated) coordinate** of Robot (r) and the location of Task (t). To account for sequential constraints, the system dynamically updates

a robot's location to the completed task's coordinates before calculating the cost for its next assigned task.

5. EXPERIMENTAL RESULTS & SIMULATION OUTPUT

Task Profiles:

- t1 -> rescue_victim, Priority 3, Location (2,8)
- t2 -> clear_debris, Priority 2, Location (7,3)
- t3 -> map_building, Priority 1, Location (1,5)
- t4 -> deliver_medicine, Priority 2, Location (4,4)

Robot Profiles:

- r1 -> mapping, Speed 3.0, Location (0,0)
- r2 -> medical, Speed 2.0, Location (5,5)
- r3 -> debris, Speed 1.0, Location (9,9)

Simulation Output:

Assignments Detected:

- Task t1 (rescue_victim) = Robot r2 (medical) [APPROVED]
- Task t2 (clear_debris) = Robot r3 (debris) [APPROVED]
- Task t4 (deliver_medicine) = Robot r2 (medical) [APPROVED]
- Task t3 (map_building) = Robot r1 (mapping) [APPROVED]

Total Plan Cost (Actual Cost): 16.00

Theoretical Optimal Cost (Lower Bound): 14.00

System Operational Efficiency (Actual/Optimal): 1.14

The system successfully satisfies all FOL constraints without any violations.

6. PERFORMANCE ANALYSIS & DISCUSSION

The simulation converged on a mathematically valid and constraint-compliant solution. Unlike Project #1, where robots computed distances strictly from their initial spawn points, this system successfully implements sequential constraints. Robots track their dynamic locations across multiple assignments.

A Theoretical Optimal Cost (Lower Bound) of 14.00 was calculated assuming an ideal scenario without sequential restrictions. The system's Actual Cost resulted in 16.00. The calculated Operational

Efficiency ratio of 1.14 (16.00 / 14.00) demonstrates a highly efficient task allocation strategy. The minor penalty of 2.00 units is a realistic and expected trade-off for physical, sequential execution where Robot r2 must travel between multiple medical tasks rather than teleporting.

This demonstrates the trade-off between strict logical safety guarantees and physical efficiency.

7. FUTURE WORK & SYSTEM ENHANCEMENTS

1. Multi-Agent Temporal Concurrency:

Future versions can support parallel task execution.

2. Constraint Relaxation:

Hard constraints may be transformed into soft constraints using utility-based optimization methods.

8. CONCLUSION

The proposed Multi-Agent System successfully solves the disaster rescue coordination problem using First-Order Logic and rational search principles.

The modular separation between the Assignment Agent and the Critic Agent provides a robust and extensible AI architecture suitable for intelligent coordination systems.